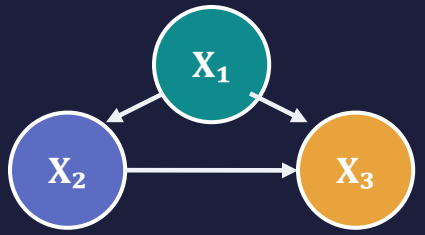
Deep End-to-End Causal Inference: Bayesian Structure Learning and Simulation-Based Treatment Effects



When does DECI's graph-recovery guarantee hold — and where does it break?

CSE 516 Probabilistic Graphical Models · Major Project M4 · Theoretical AI track

Group G5_TH_PGM ·

github.com/masoomchoksi/G5_TH_PGM (M4_G5_TH_PGM)

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From M1 scoping to M4: what we did and what we learned



Scope

Research problem, datasets, baselines, SOTA anchor (Vashishtha et al., ICLR 2025)

Theory + DECI

Reconstructed Theorem 1; ran DECI-G / DECI-S on Microsoft Causica vs NOTEARS-MLP, PC, ICA-LiNGAM

Missing data

Proof-gap audit for missingness; proposed Graph-Aware Imputation via Markov Blankets

Close the loop

Consolidated KPIs, statistics, limitations, and an honest SOTA repositioning

Causal questions need a causal graph — usually unknown

The question

"What happens to exam marks if we actively increase study hours?" — that is an intervention, not a correlation. Answering it requires knowing which variables cause which.

Why classical pipelines struggle

- Step 1: discover a graph. Step 2: estimate effects on it
- Errors in step 1 silently propagate into step 2
- Real data is mixed-type and often partially observed

Our research questions

1. Can Theorem 1 be reconstructed from its stated assumptions?
2. How does max-likelihood recovery of p(x) relate to graph recovery?
3. When does ELBO optimisation approach the ML solution?
4. Why can categorical variables make edge direction ambiguous?
5. Why does amortised imputation fall outside the proof?
6. How do these limits show up in graph recovery and ATE/CATE?

A causal question is naturally a directed graphical model

Representation

A DAG encodes direct causes as edges and gives a factorisation $p(x) = \prod p(x_i \mid pa(x_i))$

Learning

Uncertainty about the graph itself is a posterior $q(G)$ — Bayesian structure learning, not a single guess

Inference

do(X_T = a) is graph surgery: cut incoming edges, then simulate the structural equations

Uncertainty

Averaging over sampled graphs propagates structural uncertainty into ATE / CATE

Structural equations, interventions and treatment effects

Additive-noise SEM: $x_i = f_i(\text{x_pa(i)}) + z_i$, $z_i \sim \text{p_z}_i$ (independent)

Causal graph

G = (V, E) is a DAG. $X_i \rightarrow X_j$ means X_i is a direct cause of X_j. G is unknown and must be estimated from observational data.

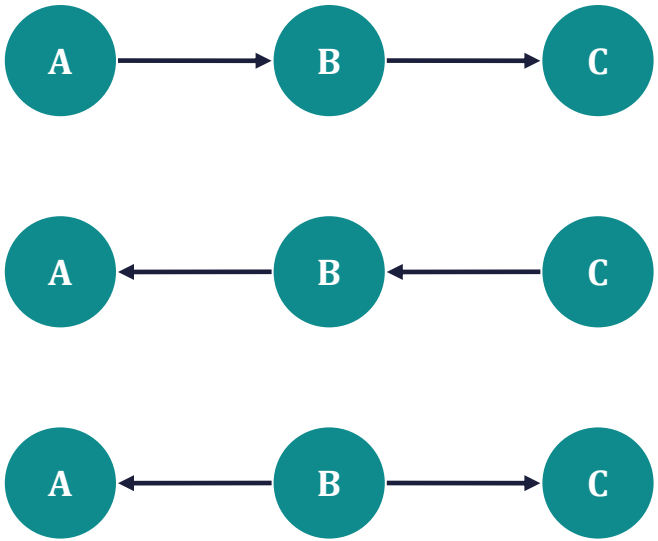
Intervention

do(X_T = a) replaces the treatment's mechanism with the constant a. The learned equations then simulate every downstream variable.

ATE / CATE

ATE = E[Y | do(X_T=a)] − E[Y | do(X_T=b)]. CATE conditions on a subgroup's covariates. DECI estimates both by simulation over graph samples.

Recovering the distribution ≠ recovering the graph



Same conditional independence (A ⊥ C | B) → one Markov equivalence class

Markov equivalence

Different DAGs can produce the same observational distribution — independence tests alone cannot tell them apart.

When is the graph identifiable?

Linear-Gaussian: generally not. Linear non-Gaussian and suitable nonlinear additive-noise models: yes.

Variational inference

DECI approximates p(G | X) with q_ϕ(G) and maximises the ELBO: data fit minus KL to the prior.

Where DECI sits among causal-discovery methods

Method	Family	Key idea	Role in our project
DECI (Geffner et al., 2022)	Bayesian, continuous optimisation	Joint graph + mechanism learning; effects by simulation	Principal model (DECI-G, DECI-S)
NOTEARS / -MLP (Zheng et al., 2018)	Continuous optimisation	Differentiable acyclicity constraint	Nonlinear baseline
PC (Spirtes et al.)	Constraint-based	Conditional-independence tests	Baseline
ICA-LiNGAM (Shimizu et al., 2006)	Linear non-Gaussian	Independent component analysis	Baseline
Additive-noise models (Peters et al., 2014)	Identifiability theory	Nonlinear ANMs can be identifiable	Underpins Theorem 1
GraN-DAG (Lachapelle et al., 2019)	Neural DAG learning	Gradient-based nonlinear DAGs	Considered, not run

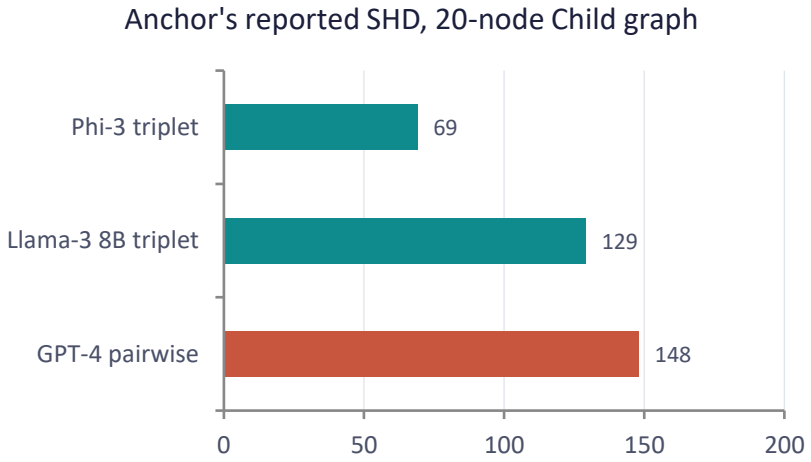
Vashishtha et al., ICLR 2025: what the anchor shows

Core idea

Imperfect experts (humans or LLMs) asked about pairwise edges confuse direct and indirect relations and create cycles. Asking for causal order — via triplet prompts — gives cycle-consistent information.

How it relates to us

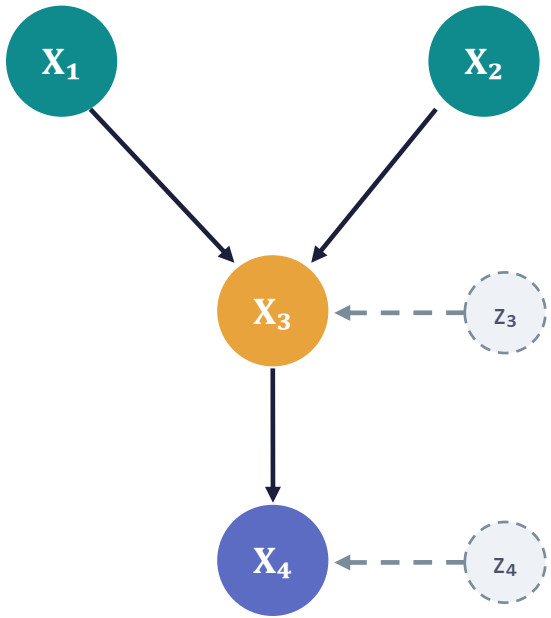
They use expert-derived order; DECI learns a graph distribution from data. Our comparison is analytical and complementary — not a numerical benchmark. Their metrics (SHD, cycles, D_top, effect error) differ from ours.



Pairwise GPT-4: >10,000 cycles · triplet: 0 cycles
Asia, CaMML effect error: 0.05 → 0.00 with LLM order (3 runs)

Numbers from the anchor paper (Tables 1, 2, 5, A7, A8), as cited in our report — context, not a like-for-like comparison.

The graphical model: nodes, edges, mechanisms and noise



$x_3 = f_3(x_1, x_2) + z_3$

$$p_\theta(x \mid G) = \prod_i p_z_i(\, x_i - f_i(x_pa_G(i)) \,)$$

- Nodes = variables; adjacency matrix A ∈ {0,1}^(D×D) encodes the graph
- Mechanisms f_i: residual MLPs with shared weights + node embeddings
- Noise: Gaussian (DECI-G) or rational-quadratic spline flow (DECI-S)
- Independent noise terms are what make direction identifiable

Conditional independencies, graph prior and graph posterior

Conditional independence

- Local Markov: each $X_i \perp$ non-descendants | parents
- In the example: $X_4 \perp \{X_1, X_2\} \mid X_3$ and $X_1 \perp X_2$
- Markov Blanket = parents + children + co-parents (used by our imputer)

Why this representation

- Directed + additive noise ⇒ identifiable under A2
- Posterior over graphs, not a point estimate
- Mean-field q is tractable but ignores edge dependencies

$$p(G) \propto \exp(-\lambda_s \|A\|^2_F - \alpha \cdot h(A) - (\rho/2) \cdot h(A)^2), \quad h(A) = \mathrm{tr}(e^{(A \circ A)}) - D$$

$h(A) \geq 0$ and $h(A) = 0$ iff A is a DAG; α, ρ follow an augmented-Lagrangian schedule ($\lambda_s = 5$).

$$q_\varphi(A_{ij}=1) = \sigma(\gamma_{ij})\sigma(\theta_{ij}), \quad q_\varphi(A_{ji}=1) = \sigma(\gamma_{ij})\sigma(-\theta_{ij})$$

ENCO parameterisation: γ = edge existence, θ = orientation. Gradients via Gumbel-softmax (temperature 0.25).

Learning graph, mechanisms and noise jointly via the ELBO

$$\text{ELBO} / N = (1/N)\cdot E_{q_\varphi(G)}[\log p_\theta(X \mid G)] - (1/N)\cdot KL(q_\varphi(G) \,\|\, p(G))$$

Data fit — how well sampled graphs + mechanisms explain the data. **Regulariser** — keeps q close to the sparse, acyclic prior.

Component	Reference setting (M2)
Structural MLP	2 hidden layers, 128 units, residual, layer norm
Noise model	Rational-quadratic spline, 8 bins (DECI-S)
Sparsity	$\lambda_s = 5$
Augmented Lagrangian (init)	$\rho = 1$, $\alpha = 0$
Optimiser	Adam, learning rate 0.01
Gradient estimator	Gumbel-softmax, $\tau = 0.25$

Why this algorithm

- Structure + parameters learned together — no two-stage error propagation
- Variational: scales where exact posterior over 2^(D(D−1)) graphs cannot
- Caveat: Adam gives no global-optimum guarantee

Effects by simulating interventions over sampled graphs

1. Sample graph $G_s \sim q_\phi(G)$
2. Mutilate: cut edges into X_T, set $X_T = a$
3. Sample noise $z^{(s,k)}$
4. Simulate $x_i = f_i(x_{pa}) + z_i$ in topological order
5. Average y over s, k; repeat for b

$$\widehat{\mathrm{ATE}} = (1/SK)\,\Sigma_s\,\Sigma_k\, \hat{y}^{(s,k)}(a) - (1/SK)\,\Sigma_s\,\Sigma_k\, \hat{y}^{(s,k)}(b)$$

Configurations

ATE: S = 1,000 graphs, K = 2 noise draws. CATE: 10 graphs, 10,000 draws, ridge regression on 3,000 random Fourier features.

Computational trade-off

- Exact inference over all graphs is intractable; nonlinear flows rule out closed-form message passing
- Simulation cost $\propto S \times K$ forward passes through D mechanisms
- Price: Monte Carlo variance + dependence on posterior quality
- Error sources: MC noise, wrong graph, mis-specified f_i, CATE surrogate

Our assumption audit: five stated, five implicit

Stated in DECI

A1	Minimality
A2	Structural identifiability
A3	Correct specification
A4	Causal sufficiency
A5	Regularity (finite log-lik.)

Implicit — identified in our audit

A6	Uniform convergence / argmax
A7	Optimum representable by q family
A8	Graph support; cyclic-graph prob.
A9	Global optimum vs practical optimiser
A10	Bounded prior hyperparameters

Theorem 1 (informal): under these assumptions, as N → ∞ the ELBO maximiser recovers the true graph.

Five steps from likelihood to graph recovery

1	**Valid DAG likelihood**	Topological order ⇒ unit lower-triangular Jacobian, det = 1, so p(x|G) factorises. Not valid for cyclic samples → we track the cyclic-sample rate.
2	**Prior & acyclicity**	h(G) is finite, non-negative and zero exactly for DAGs.
3	**ML recovers p, not G**	Gibbs: E[log p_θ] − E[log p*] = −KL(p* ‖ p_θ) ≤ 0. Under A3 the maximiser equals p* — but many graphs may give p*.
4	**Identifiability closes the gap**	Under A2, only the true (G, θ) gives p*. So the ML solution has the true graph.
5	**ELBO → ML**	Finite graph space + p(G) ≥ ε ⇒ KL(q‖p) ≤ log(1/ε) = O(1), so KL/N → 0. Needs A6–A9 for the optimiser too.

Where the proof breaks: categorical and missing data

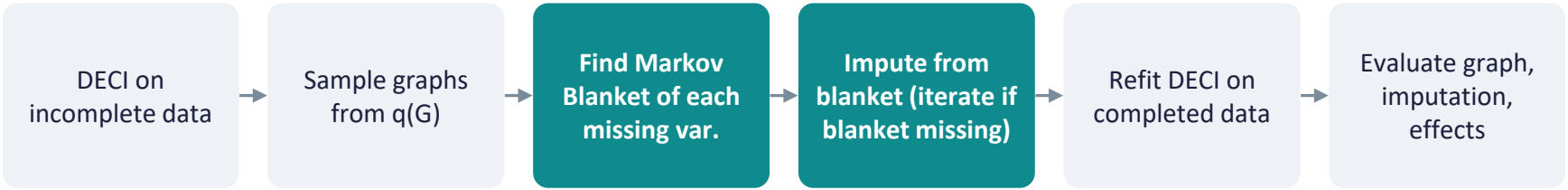
Boundary 1 — categorical (Proposition 2)

- Two binary variables: $p(x_1)p(x_2|x_1) = p(x_2)p(x_1|x_2)$
- Both directions have 3 free parameters and fit every joint exactly
- A2 fails; equal sparsity cost ⇒ the prior can't break the tie
- Prediction: adjacency recoverable, direction uncertain → evaluate adjacency and orientation separately
- Status: proposition from analysis, not verified empirically

Boundary 2 — missing data (Proposition 3)

- Each missing entry becomes a per-observation latent z_n with its own q_n
- Gap = Σ_n KL(q_n ‖ p(z_n | x_n^obs, G)) — a sum of N terms
- If each term ≥ δ > 0, gap ≈ Nδ, so gap / N does not vanish
- Step 5 (KL/N → 0) breaks; Gaussian, graph-agnostic imputer may mismatch
- No missing-data recovery theorem is established

Graph-Aware Imputation: impute from the Markov Blanket



Variant	Description	Role
A	Original DECI (graph-agnostic amortised imputation)	Required control
B	Mean / median fill, then refit	Basic control
C	Agnostic MLP on all other variables	Separates structure from capacity
D	Graph-aware deterministic (MLP on Markov Blanket)	First prototype
E	Graph-aware stochastic (samples from predicted dist.)	Preferred proposal
F	Oracle true graph	Diagnostic only

Honest framing

- Adds a structural inductive bias, no new information
- Two-stage: does not change the ELBO
- MCAR at 0 / 10 / 30 / 50%

Datasets, protocol and KPIs

Data & methods

- Synthetic ER DAGs (8–32 nodes), nonlinear ANM with spline-warped noise
- CSuite: lingauss, linexp, nonlingauss, nonlin_simpson
- DECI-G, DECI-S vs NOTEARS-MLP, PC, ICA-LiNGAM

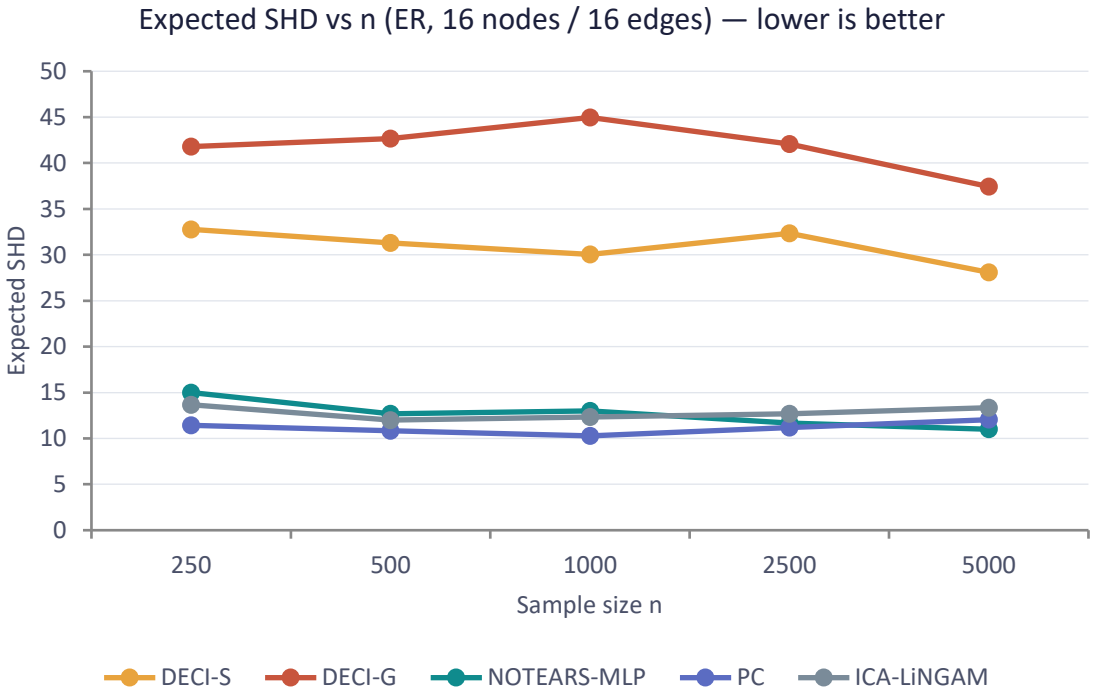
Protocol

- Matched data and splits; documented seeds
- 3 seeds (2 in DECI-S validation arm)
- Same missingness mask per seed and rate
- Missing runs reported, not dropped

Stage	KPIs
Representation	SHD ↓, adjacency F1 ↑, orientation F1 ↑, posterior mass on true graph, cyclic-sample rate
Learning	ELBO, held-out NLL, log-loss
Inference	ATE RMSE ↓, CATE RMSE, runtime, MC variability
Missing data	Imputation RMSE / MAE vs mean-fill
Probability quality	NLL, log-loss, Brier, ECE

Planned in M1 but without reported results: SynTReN, Sachs, IHDP, Twins; E2, E3, E6, E8–E10.

More data did not reliably help DECI; baselines won on SHD



28.1 ± 7.9

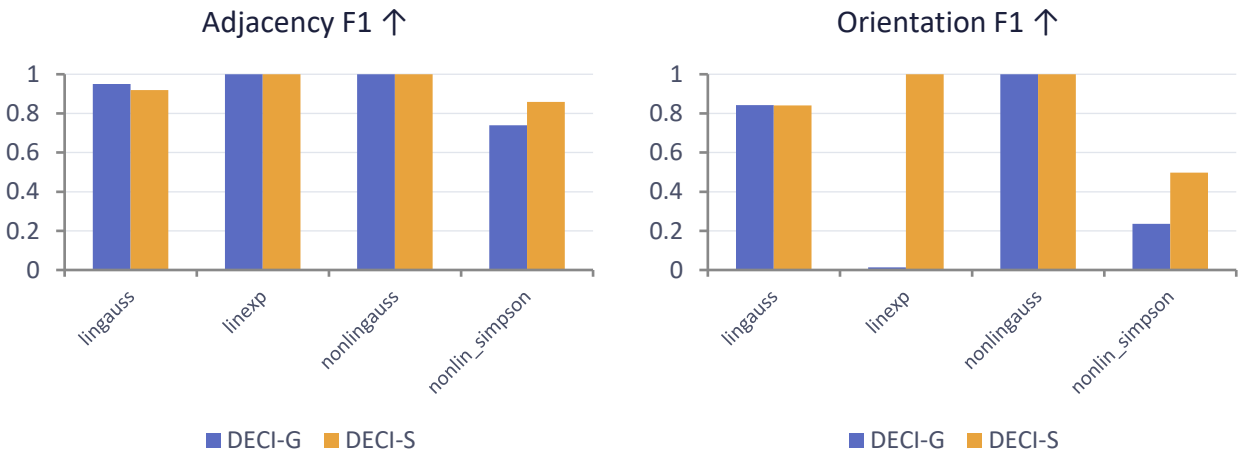
DECI-S at n = 5000 — best DECI result, still above every baseline

10.3 ± 2.2

PC at n = 1000 — lowest SHD in the table

Prediction "more data ⇒ better DECI" is only partially supported: the asymptotic argument says nothing about finite-sample Adam training.

Perfect adjacency does not imply correct orientation



ATE RMSE ↓	G (ours)	S (ours)	G pub.	S pub.
lingauss	0.136	0.169	0.120	0.149
linexp	**0.989**	0.014	1.029	0.022
nonlingauss	0.005	0.016	0.042	0.043
nonlin_simpson	1.855	0.889	1.995	1.997

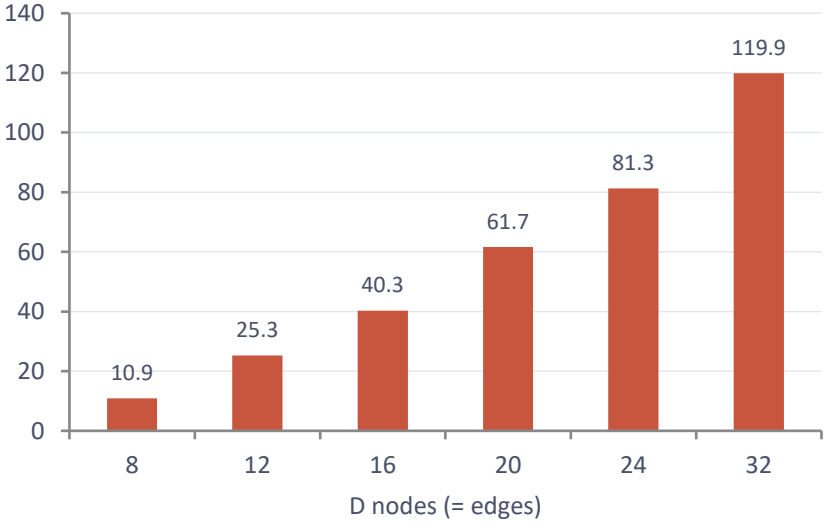
1.0 → 0.013

linexp, DECI-G: adjacency F1 → orientation F1

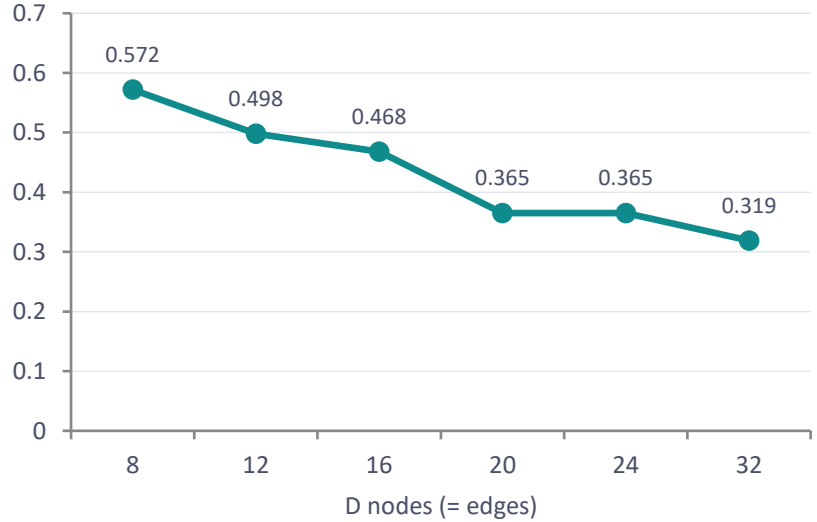
Gaussian noise cannot model exponential noise, so DECI-G finds the skeleton but points edges the wrong way — and its ATE is badly wrong (0.989). DECI-S, with a flexible noise flow, gets both right.

Recovery degrades as graphs grow (DECI-G, n = 1000)

Expected SHD ↓ by node / edge count

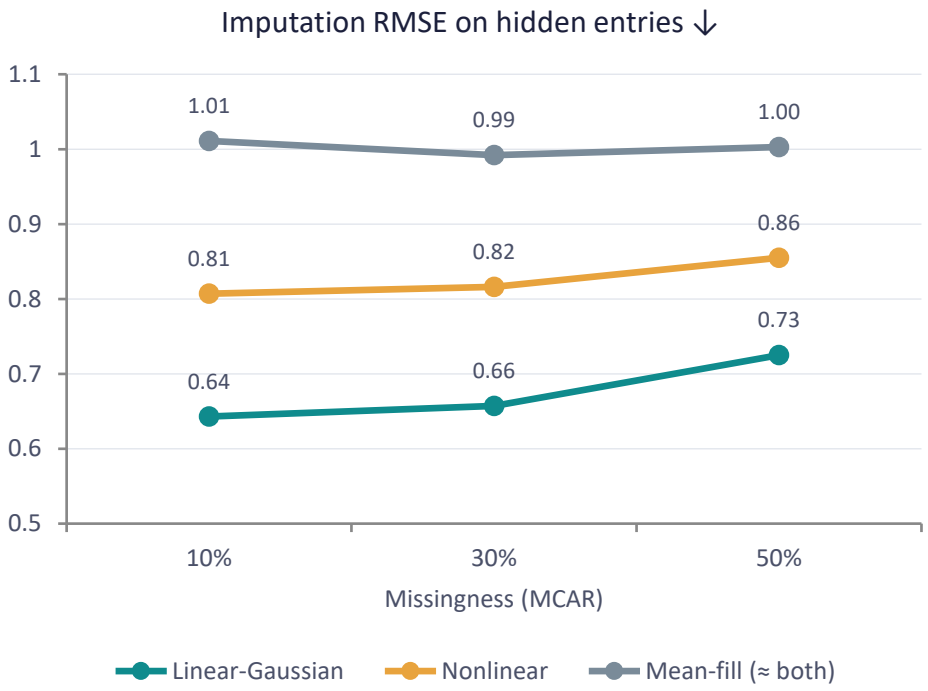


Adjacency F1 ↑



Cyclic-sample rate 0.000 in every DECI-G setting. DECI-S check: D = 8 SHD 9.67 ± 2.09 (1.6 h); D = 24 SHD 69.78 ± 0.77 (4.8 h vs 0.75 h for DECI-G).

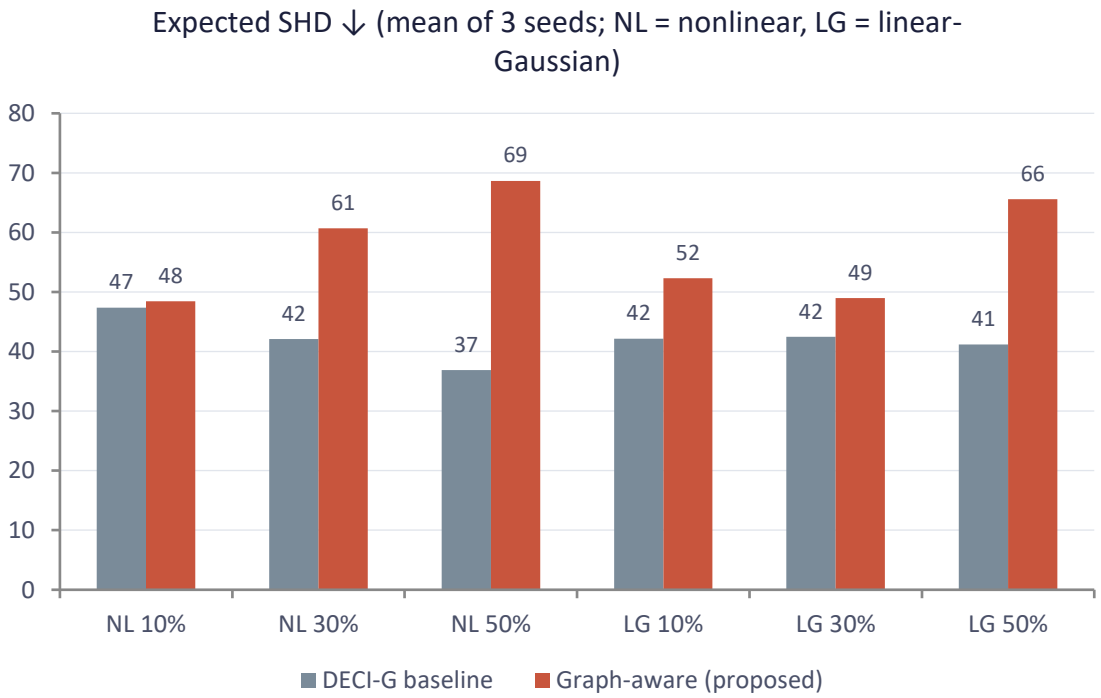
Under MCAR missingness, imputation error rises steadily



Regime	Miss.	Exp. SHD	Adj. F1	Orient. F1
Lin-Gauss	0%	53.03 ± 2.76	0.337	0.089
Lin-Gauss	10%	42.17 ± 7.14	0.449	0.215
Lin-Gauss	50%	41.21 ± 3.78	0.474	0.180
Nonlinear	0%	40.26 ± 3.01	0.468	0.216
Nonlinear	10%	47.36 ± 3.88	0.380	0.126
Nonlinear	50%	36.87 ± 4.54	0.484	0.237

Graph metrics are non-monotonic in missingness — this does not show that more missingness helps.

Graph-Aware Imputation made DECI-G's SHD worse, not better



6 / 6

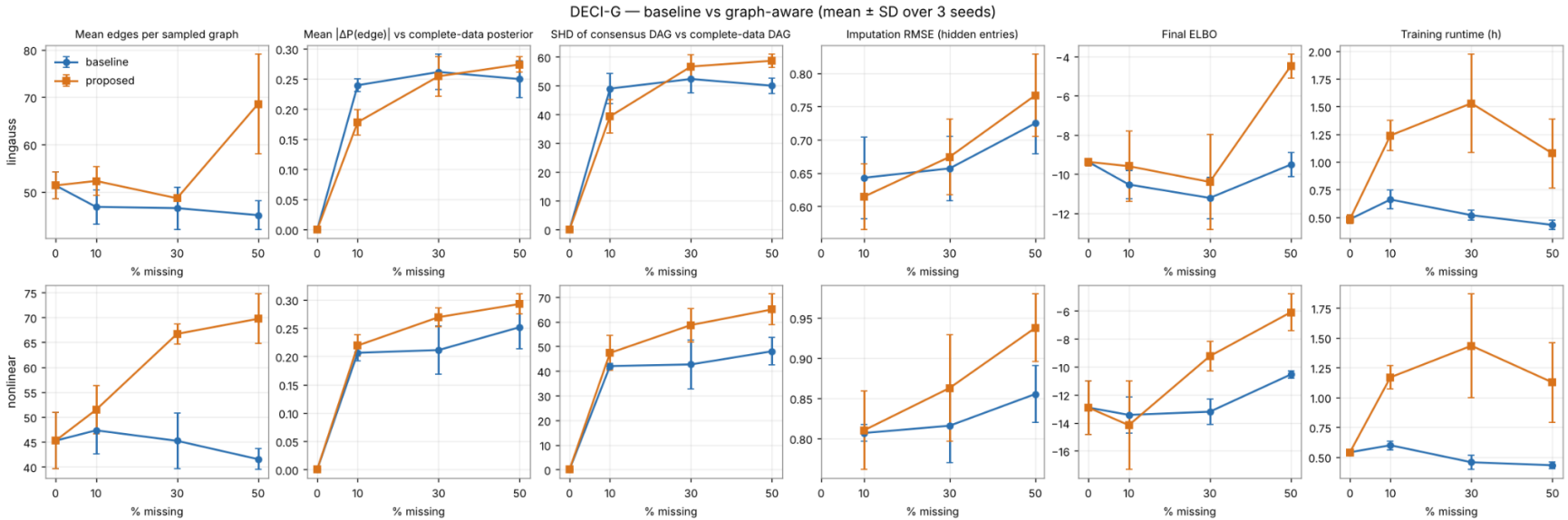
conditions where SHD got worse with graph-aware imputation

+31.8

SHD gap at nonlinear 50% — the gap grows with missingness

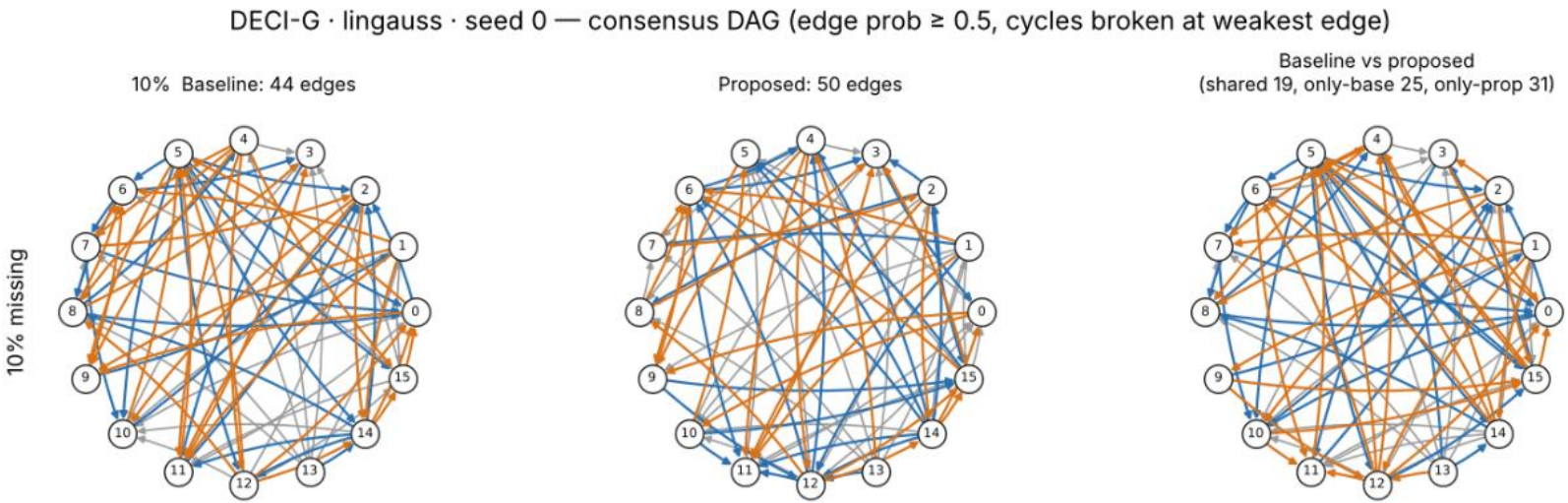
Adj. F1 lower in 5 of 6; imputation RMSE similar (better only LG 10%: 0.614 vs 0.643, within 1 SD). Likely cause (untested): first-stage graph errors (adj. F1 ≈ 0.4) feed the Markov-Blanket imputer.

Baseline vs graph-aware on six diagnostics (DECI-G)



Proposed graphs are denser (mean edges 45 → 70 in nonlinear), drift further from the complete-data posterior, and train ~2–3× longer. **A higher final ELBO did not mean a better graph — fit is not identification.**

Consensus DAGs at 10% missingness (DECI-G, lingauss)



Edge colours (panels 1–2)

Grey: also in complete-data graph · Blue: extra edge · Orange: edge lost

Panel 3: baseline vs proposed

Only 19 shared edges; 25 baseline-only, 31 proposed-only — the two pipelines disagree substantially

Is the SHD increase significant? 95% CIs and Welch tests

Condition	Baseline SHD [95% CI]	Graph-aware SHD [95% CI]	Δ SHD	Welch p	Cohen's d	Holm (α=0.05)
Nonlinear 10%	47.4 [37.7, 57.0]	48.5 [39.8, 57.1]	+1.1	0.736	0.30	n.s.
Nonlinear 30%	42.1 [32.3, 51.9]	60.7 [53.4, 68.0]	+18.6	0.004	5.36	worse
Nonlinear 50%	36.9 [25.6, 48.1]	68.7 [56.4, 80.9]	+31.8	0.001	6.71	worse
Lin-Gauss 10%	42.2 [24.4, 59.9]	52.3 [41.0, 63.6]	+10.1	0.119	1.69	n.s.
Lin-Gauss 30%	42.5 [26.3, 58.6]	49.0 [44.2, 53.8]	+6.5	0.218	1.36	n.s.
Lin-Gauss 50%	41.2 [31.8, 50.6]	65.6 [43.8, 87.3]	+24.4	0.026	3.61	n.s. after Holm

How these were computed

From reported mean ± SD with n = 3: CI = mean ± $t_{0.975,2}$ · SD/√3 ($t = 4.30$). Unpaired Welch test — conservative, since seeds and masks are matched.

Still to complete

Paired tests on raw per-seed runs; NLL, log-loss, Brier and ECE; verified seed counts (E1). Significance ≠ proof of a theoretical claim.

Why the results came out the way they did

Theory vs practice

Theorem 1 is asymptotic and assumes the global optimum. Finite-sample Adam training need not be monotone in n (E1).

Mis-specified noise breaks direction

linexp: Gaussian DECI-G gets the skeleton but not orientation — A3 fails, the identifying asymmetry is lost.

Two-stage error propagation

Graph-aware imputation inherits first-stage graph errors (adj. F1 ≈ 0.4), so wrong blankets → wrong imputations → denser graphs.

Fit ≠ identification

Higher final ELBO with worse SHD — exactly the gap between Step 3 and Step 4 of the proof.

Identify → Understand → Compare → Validate → Reposition

Dimension	Vashishtha et al. (ICLR 2025)	This project
Information source	Imperfect experts (human / LLM) on causal order	Observational data, fully or partially observed
Core problem	Pairwise judgments confuse direct / indirect, form cycles	When DECI's recovery guarantee holds and where it breaks
Evaluation	Cycles, causal-order accuracy, effect error	SHD, adj./orient. F1, ATE RMSE, imputation RMSE
Shared benchmark?	No	No

M1 expectation

A missing-data extension of DECI that could improve recovery and be positioned against the anchor.

Where we are strong

Explicit A1–A10 audit; adjacency vs orientation separation; categorical and missing-data boundaries.

Repositioned (M4)

Analytical, complementary work. We do not outperform the anchor; our proposed method did not beat its control.

What our work does not do, and where it breaks

Theoretical

- Depends on A2–A5; categorical result is a proposition, not a theorem
- ELBO convergence needs A6–A9
- Adam: no global guarantee
- Mean-field q ignores edge dependencies

Missing data

- Only MCAR — no MAR / MNAR
- Graph errors propagate into imputation
- Two-stage: ELBO unchanged, no imputation uncertainty in refit
- Gaussian-family mismatch remains

Experimental

- Only part of the M1 plan has results
- Variants B–F and ATE/CATE under missingness not run
- Paired tests, Brier / ECE, NLL missing
- Few seeds (2–3); synthetic data only

The next research steps

1. Run the full six-variant comparison (A–F) with shared masks and seeds
2. Multiple graph samples + multiple imputation to carry uncertainty into the refit
3. Move imputation inside the variational objective instead of two-stage
4. Identifiability under MAR / MNAR missingness
5. Empirically test the categorical orientation proposition
6. Causal-order priors (the anchor's idea) as DECI graph priors
7. Calibration (Brier / ECE), ATE / CATE under missingness, real data (Sachs, IHDP)

What we learned

- Recovering the distribution is not recovering the graph — identifiability is what bridges them.
- Adjacency can be perfect while orientation fails (1.000 vs 0.013): evaluate both.
- A guarantee at the global optimum says little about finite-sample Adam training.
- Our Graph-Aware Imputation did not help: structure-based methods are only as good as the structure.
- Next step: put imputation inside the variational objective, under MAR / MNAR.

Key references

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